

High-Precision Micro-Hole Depth Measurement Method and System Based on Computer Vision and Deep Learning Error Compensation

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Abstract—To address the issues of high cost and the failure of focus measure in existing micro-hole depth measurement equipment under uncontrolled physical environments, this paper proposes a novel micro-hole measurement method based on machine vision and deep learning error compensation. First, a Weighted Mean-Multiscale Aggregation-Normalized Joint Laplacian Variance (WMN-LV) algorithm is developed to effectively overcome the interference of complex illumination and environmental noise, realizing precise focusing and preliminary feature extraction of micro-hole samples. Secondly, aiming at the non-linear errors caused by physical limits, a multi-layer deep learning regression network is introduced for error compensation. Through data augmentation and regularization strategies, small-sample overfitting is effectively avoided, and the relative measurement error of physical micro-hole samples is further reduced from 5% to within 3%. Finally, a low-cost optical precision measurement system is independently built as an experimental verification platform. Experimental results show that, compared with traditional algorithms such as Tenengrad and Brenner, the proposed WMN-LV algorithm combined with the deep learning compensation mechanism has significant advantages in peak positioning accuracy and robustness, providing an efficient algorithmic solution for micro-size non-destructive testing in complex environments.

Keywords—Micro-hole depth measurement, Computer vision, Laplacian variance, Deep learning, Error compensation.

I. INTRODUCTION

Micro-hole depth measurement is a core supporting technology in high-end manufacturing, such as aerospace and microelectronics^{[1], [2]}. While traditional methods like laser interferometry^[3] and confocal microscopy^[4] offer high precision, they suffer from high equipment costs and extreme sensitivity to physical testing environments. Consequently, low-cost automated vision-based measurement has become a research hotspot. However, in uncontrolled physical environments, existing focus measure algorithms (e.g., Tenengrad, Brenner, and SML) exhibit weak noise resistance, blurred peaks, and low positioning accuracy^[5], making high-confidence measurement difficult.

To break through these visual algorithm bottlenecks and physical accuracy limits, this paper proposes a novel joint Laplacian variance algorithm (WMN-LV) combined with deep learning error compensation. Unlike traditional edge-based

methods that strictly rely on local high-frequency gradients and are highly susceptible to noise, the proposed approach synergizes bilateral filtering for edge-preserving noise reduction and multi-scale LoG (Laplacian of Gaussian) operator aggregation to capture structural focus features across varying degrees of blur, thereby fundamentally enhancing the global stability of the focus measure.

The main contributions are:

1. A WMN-LV focus measure algorithm is proposed, achieving sharp peak positioning and highly robust focusing in non-ideal lighting environments.
2. A multi-layer deep learning regression model is constructed. By introducing a weighted fusion mechanism, systematic physical errors are successfully controlled within 3%.
3. A low-cost measurement verification platform is independently built. Experimental results verify that the proposed method comprehensively outperforms existing classical algorithms, providing a reliable alternative for micro-size non-destructive testing.

II. WORK PRINCIPLE AND OVERALL DESIGN

Based on the spatial invariance of a convex lens's focal length, the geometric depth of the blind hole corresponds to the absolute axial displacement difference when focusing on the top and bottom surfaces: $z_{depth} = |z_{top} - z_{bottom}|$, where z_{top} and z_{bottom} are the Z-axis readings at the respective focal planes. The overall methodology encompasses: Generating computer-simulated z-stack image sequences using the Point Spread Function (PSF); Developing a robust WMN-LV autofocus algorithm; Integrating a low-cost optical measurement system for physical verification.

III. MICRO-HOLE STRUCTURE SIMULATION AND ALGORITHM DEVELOPMENT

A. Generation of Clear Base Images of Samples

To generate computer-simulated samples highly consistent with real sample morphologies, a multi-decision-variable image generation method is proposed. For any point (x, y) in the image, the finally generated image $I_{final}(x, y)$ is denoted as:

$$I_{final}(x, y) = (1 - M) \cdot B(x, y) + M \cdot (\gamma \cdot B(x, y) + I(x, y)) \quad (1)$$

where parameter M is the micro-hole region determination factor, used to distinguish the interior and exterior regions of the micro-hole in the image; the noise function $B(x, y)$ is the texture generated by combining a random noise algorithm with low-pass filtering; and $\gamma \in [0.1, 0.3]$ is the background brightness parameter inside the hole. $I(x, y)$ is a radial attenuation function used to simulate the illumination attenuation inside the hole, expressed as:

$$I(x, y) = A \cdot \exp(-\beta \cdot r(x, y)^2) + \eta(x, y) \quad (2)$$

where r is the normalized radius, and $\eta(x, y)$ is a zero-mean random field used to create fine textures, with Gaussian noise serving as the noise source during generation. A is the amplitude controlling the central contrast, thereby adjusting the brightness and darkness of the image inside the hole. β controls the steepness of the radial attenuation, facilitating the measurement at the center of the bottom surface of the simulated sample.

It is worth noting that the micro-hole region determination factor M is not a static spatial mask. In practical physical measurements, M is dynamically segmented using adaptive thresholding (e.g., Otsu's method). Furthermore, in hardware instrument set design, the employment of a high-brightness ring light source effectively suppresses severe ambient illumination interference, ensuring the robust extraction of hole boundaries.

A Python computer program was developed to generate five different micro-hole base images, whose geometric parameters are detailed in Table I. Figure. 1(a) presents the real sample image captured by an industrial camera, while Figure. 1(b)–(f) show the computer-simulated images. It is evident that the computer-simulated images are highly consistent with the actual images captured by the industrial camera, laying a solid foundation for subsequent simulation experiments.

TABLE I. GEOMETRIC PARAMETERS OF FIVE SIMULATED SAMPLES

Simulation sample number	Depth	Diameter	Ratio of depth to diameter
1	1200	200	6.00
2	1000	120	8.33
3	500	180	2.78
4	800	300	2.67
5	900	500	1.80

^a. The units for depth and diameter are micrometers.

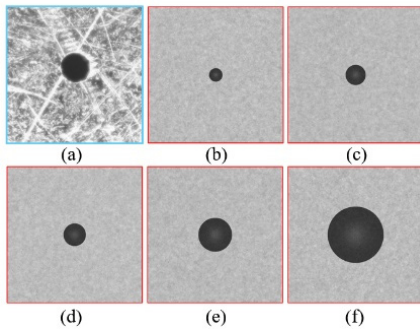


Figure 1. Real samples and simulation samples images

B. Generation of z-stack Images Using Point Spread Function (PSF)

For each sample, a z-stack image sequence is generated along the Z-axis at fixed steps to simulate the imaging effects at different focal planes during the focusing process. The Point Spread Function (PSF)^[6] is introduced and applied to the base image $I_{final}(x, y)$ to generate this sequence of defocused images, denoted as I_{new} . The expression is defined as:

$$I_{new}(x, y; z) = (I_{final} * \text{PSF}(\cdot; \sigma))(x, y) + n(x, y) \quad (3)$$

where σ is the blur scale (in pixels) related to the defocus amount Δz , $n(x, y)$ represents Gaussian noise, and Z after the semicolon denotes the position parameter. This equation is utilized to generate defocused blurred images diffused by the PSF.

It should be noted that while the blur scale is expressed in pixels for algorithmic simulation, a pre-calibrated spatial resolution factor (e.g., 1 pixel $\approx$ 1.25 μm) is applied during physical experiments to precisely convert pixel-level disparities into physical micrometer (μm) measurements, as evaluated in Table I and subsequent sections.

A Python computer program was developed to generate these images based on the clear base images produced previously (as shown in Figure. 1). Taking one specific sample as an example, the resulting z-stack image sequence is illustrated in Figure. 2. This sequence is constructed on the basis of the clear top-surface and bottom-surface base images of the micro-hole, utilizing the PSF to simulate imaging effects under varying out-of-focus states. The color bar at the top indicates the depth variation from focusing on the top surface (blue end) to the bottom surface (red end).

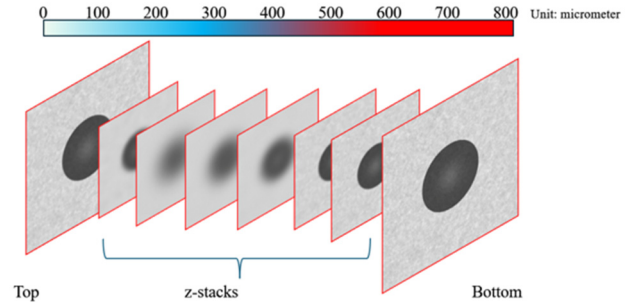


Figure 2. The z-stack image sequence of a single simulated sample.

C. Weighted Mean-Multiscale Aggregation-Normalized Joint Laplacian Variance (WMN-LV) Algorithm

1) Algorithm Principle:

To address the limitations of traditional focus measure algorithms such as Tenengrad, Brenner, and SML, a Weighted Mean-Multiscale Aggregation-Normalized Joint Laplacian Variance (WMN-LV) method is proposed. The specific principles are detailed as follows:

a) Pre-filtering processing: The original grayscale image is initially filtered to eliminate high-frequency noise. Based on the computation of a 4-neighborhood discrete operator^[7], bilateral filtering is applied to the grayscale image to enhance accuracy

and mitigate noise interference before calculating the Laplacian response L [8]–[10]:

$$L = \mathcal{L}[I](x, y) = I(x+1, y) + I(x-1, y) + I(x, y+1) + I(x, y-1) - 4I(x, y) \quad (4)$$

b) Calculation of weighted variance: By utilizing the determination parameter M , the weighted mean μ_w and weighted variance Var_w are computed to accentuate the morphological features of clarity inside the micro-hole:

$$\mu_w = \frac{\sum_{x,y} M(x, y) L(x, y)}{\sum_{x,y} M(x, y)} \quad (5)$$

$$\text{Var}_w(L) = \frac{\sum_{x,y} M(x, y) (L(x, y) - \mu_w)^2}{\sum_{x,y} M(x, y)} \quad (6)$$

c) Introduction of multi-scale aggregation and normalization: The Laplacian of Gaussian (LoG) operator [11], is utilized for aggregation to derive the joint variance S :

$$S = \sum_k w_k \text{Var}(\text{LoG}_{\sigma_k}) \quad (7)$$

where w_k is the weight coefficient corresponding to each scale, and σ_k represents the set of scales (in pixels) [12]. To facilitate cross-sample comparisons and attenuate the influence of exposure variations, the scoring results must be normalized:

$$\text{Score}_{\text{norm}} = \frac{\text{Var}_w(L)}{(E_w[I])^2 + \varepsilon} \quad (8)$$

where $\text{Score}_{\text{norm}}$ represents the final focus measure score [13], $E_w[I]$ is the mean pixel value within the hole, and ε (set to 1×10^{-6}) is a small constant introduced to prevent zero-division errors while maintaining normalization stability.

2) Algorithm Comparison and Validation:

To verify the feasibility of the WMN-LV algorithm, simulation experiments were first conducted utilizing the computer-generated samples, and synchronously compared with three classical algorithms (Tenengrad, Brenner, and SML). The depth measurement results of the four algorithms on the simulated samples are presented in Figure 3.

As observed from the evaluation curves, although the Tenengrad algorithm exhibits sharp peaks, it remains highly unstable under external noise; in scenarios without a light-proof enclosure, ambient light significantly compromises its precision. The Brenner algorithm demonstrates high computational efficiency but lacks adequate sensitivity for small-aperture samples, restricting its application to large-scale structural measurements. Furthermore, the SML algorithm yields blurred peak positioning and lowest measurement accuracy.

The simulation experiments substantiate that the proposed WMN-LV algorithm possesses significant advantages, notably a stable and low baseline coupled with sharp peak responses. Its stability and accuracy under complex working conditions

substantially outperform those of traditional methods, thereby providing a highly reliable algorithmic foundation for subsequent physical measurements.

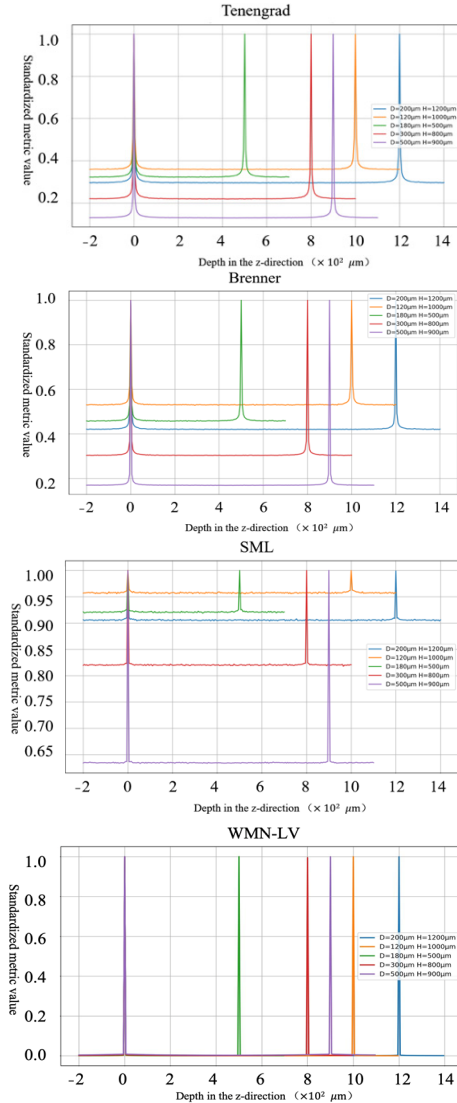


Figure 3. Depth measurement results of the simulated samples using four different algorithms.

In terms of computational efficiency, the WMN-LV algorithm achieves a time complexity of $O(W \times H)$ per frame, where W and H represent the image dimensions. During real-time processing on a standard PC (Intel Core i5, 16GB RAM), the average execution time per frame is restricted to approximately 20 ms, fully satisfying the requirements for dynamic real-time focus tracking.

IV. SYSTEM IMPLEMENTATION AND EXPERIMENTAL VERIFICATION

A. Structural Design of the Optical Measurement System

The measuring instrument adopts a modular design, comprising an optical stand, an industrial camera, a microscopic magnification lens, a high-brightness ring light source, and a precision three-axis stage. The structure is unshielded on all sides, facilitating experimental observation and manual

operation. The 3D conceptual model of the mechanical structure is shown in Figure. 4(a), and the complete apparatus connected to a computer is depicted in Figure. 4(b). The equipment is easy to assemble, and the industrial camera and microscopic lens can be replaced according to various sample requirements. During measurement, the sample is placed on the stage, and the precision three-axis stage is adjusted to position the sample at the center of the camera's field of view.

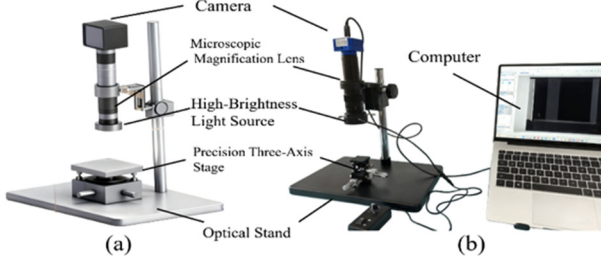


Figure 4. The 3D model and the complete apparatus of the measurement system.

B. Design of the Measurement Software

Two sets of visualization software were developed:

- *Image processing software*: Incorporating the WMN-LV algorithm, it captures images in real time and generates a dynamic focus curve, synchronously outputting a focus score. When the normalized focus score $Score_{norm}$ reaches a predefined empirical threshold (set to $Score_{norm} \geq 0.95$), the target box turns green to indicate precise focusing, and the Z-axis reading is simultaneously recorded.
- *Data processing software*: It supports the one-click import of multiple sets of measurement data and automatically generates intuitive schematic diagrams and error analysis reports.

C. Measurement Verification of Physical Samples

To verify the reliability of the aforementioned design and algorithm, as well as the actual measurement accuracy, three physical samples were selected for testing. The samples are rectangular aluminum plates with micro-holes drilled by precision drill bits, as shown in Figure. 5(a). They were tested by a third-party testing institution using a high-precision confocal microscope to provide standard reference values. The image captured by the computer is shown in Figure. 5(b), and the measurement process is illustrated in Figure. 5(c).

Before error compensation, the relative measurement error exhibits a positive correlation with the micro-hole aspect ratio. Specifically, as the aspect ratio increases from 3:1 to approximately 10:1, the initial visual measurement error escalates from around 2.1% to near 4.8% due to intensified internal wall reflections.

The final data summary is presented in Table II. This table records the measured depths obtained by our instrument and the high-precision detection values (denoted as the standard reference depth), alongside the calculated relative errors. It is evident that the relative errors using the proposed method are all controlled within 5%, successfully meeting the requirements for low-cost and high-precision measurement.

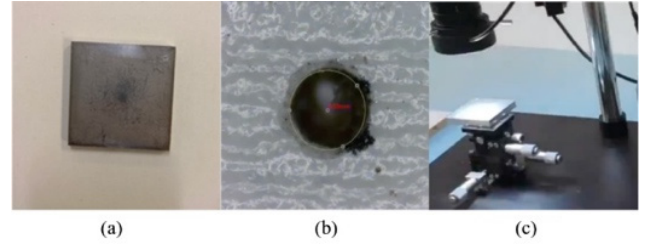


Figure 5. The physical sample and its measurement process.

TABLE II. MEASUREMENT RESULTS AND RELATIVE ERRORS OF PHYSICAL SAMPLES.

Sample No.	Standard Reference Depth (μm)	Measured Depth (μm)	Relative Error (%)
1	1264	1202 $\pm$ 5	4.90
2	1103	1150 $\pm$ 5	4.26
3	1344	1301 $\pm$ 5	3.20

V. DEEP LEARNING ERROR COMPENSATION AND ACCURACY ENHANCEMENT

A. Multi-layer Regression Network Structure and Optimization

To further overcome systematic non-linear errors caused by complex illumination and high-reflectivity specimens, a multi-layer deep learning regression network is introduced. The network employs the hyperbolic tangent (Tanh) non-linear activation function to extract micro-hole image features and achieve high-dimensional non-linear mapping, avoiding linear model degradation. Mean Squared Error (MSE) is utilized as the training loss function to precisely fit the relative errors of physical dimensions.

To prevent overfitting due to the limited physical samples, data augmentation (e.g., random rotation, Gaussian noise) is executed, expanding the dataset a hundredfold. Furthermore, an L2 regularization penalty and a dropout mechanism are integrated into the hidden layers^{[14], [15]}. The details are shown in Figure 6.

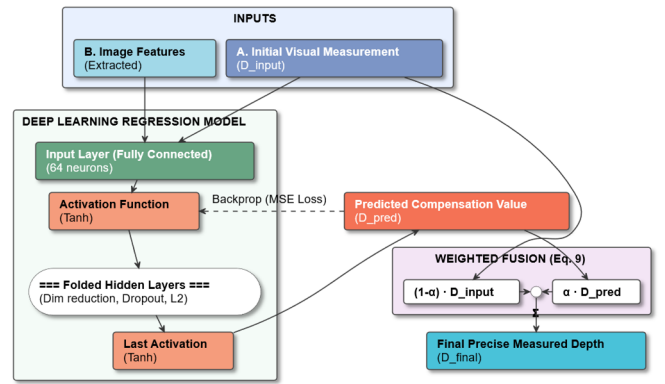


Figure 6. Topology and training framework of the proposed deep learning regression network.

During the backpropagation process, the Adam optimizer is utilized to update the network parameters, including the adaptive fusion weight α , with an initial learning rate set to 0.001. The training process mathematically converges when the validation loss changes by less than 1×10^{-4} for 10 consecutive epochs, or when it reaches a maximum of 200 epochs.

B. Weighted Fusion Mechanism and Experimental Evaluation

Ultimately, to balance the baseline stability of the initial visual measurement with the dynamic prediction accuracy of the neural network, a weighted fusion error compensation mechanism is introduced. The final measured depth D_{final} is formulated as:

$$D_{final} = \alpha \cdot D_{pred} + (1 - \alpha) \cdot D_{input} \quad (9)$$

where D_{pred} is the model predicted value from the deep learning network, D_{input} represents the initial visual measurement derived from the WMN-LV algorithm, and α signifies the fusion weight parameter learned adaptively through backpropagation.

The measurement results and relative errors of the physical samples after model fusion are compiled in Table III. Experimental results demonstrate that, after applying the deep learning error compensation, the relative measurement errors of the physical micro-hole samples are successfully suppressed from within 5% to further within 3%, minimizing the systematic physical deviations under complex working conditions.

TABLE III. MEASUREMENT RESULTS AND RELATIVE ERRORS AFTER MODEL FUSION

Sample No.	Standard Value (μm)	Model Predicted Value (μm)	After Fusion (μm)	Relative Error (%)
1	1264	1244 $\pm$ 5	1233 $\pm$ 5	2.45
2	1103	1129 $\pm$ 5	1135 $\pm$ 5	2.90
3	1344	1337 $\pm$ 5	1327 $\pm$ 5	1.26

VI. CONCLUSION

The proposed WMN-LV algorithm and its supporting physical platform for micro-hole depth measurement exhibit core advantages of low cost, high precision, and ease of operation. The total equipment cost is restricted to within 1,000 CNY. The relative measurement error is tightly controlled within 5%, and is further reduced to less than 3% after integrating the deep learning fusion mechanism.

This algorithm and verification platform possess broad application prospects. They are expected to provide a low-cost precision measurement solution for baseline micro-hole inspection in fields such as aerospace and microelectronics. Furthermore, through the deep integration of computer vision algorithms and photoelectric imaging, this work substantially enriches the technical application and analytical testing of computer vision algorithms at the micro-nano scale. Future endeavors will focus on optimizing the algorithm's adaptability under more extreme operating conditions, exploring lightweight embedded deployment to improve portability, and continuously iterating the high-performance algorithmic architecture to provide efficient solutions for automated non-destructive testing.

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